

P8131 Homework 10

Yongyan Liu (yl6107)

2026-04-25

Problem 1

Suppose we have the following data:

- **Group 1:** 4, 12+, 15, 21+, 23
- **Group 2:** 2, 6+, 8+, 10, 19

(+ denotes right-censoring.) Test $H_0: h_1(t) = h_2(t) \forall t$ versus $H_1: h_1(t) \neq h_2(t)$ for some t , using the log-rank test.

Hypotheses (Lec 22, slide 6).

$$H_0: S_1(t) = S_2(t) \text{ for all } t \quad \left(\Leftrightarrow h_1(t) = h_2(t) \right) \quad \text{vs.} \quad H_1: S_1(t) \neq S_2(t) \text{ for some } t.$$

At each *uncensored* pooled event time t_i , expected count and hypergeometric variance under H_0 are

$$E(d_{1,i}) = \frac{m_{D,i} n_{1,i}}{n_i}, \quad \text{Var}(d_{1,i}) = \frac{n_{1,i} n_{2,i} m_{D,i} m_{\bar{D},i}}{n_i^2 (n_i - 1)}.$$

The pooled uncensored event times are $\{2, 4, 10, 15, 19, 23\}$. The per-event-time table is

t_i	$n_{1,i}$	$n_{2,i}$	n_i	$d_{1,i}$	$d_{2,i}$	$m_{D,i}$	$E(d_{1,i})$	$\text{Var}(d_{1,i})$
2	5	5	10	0	1	1	0.5000	0.2500
4	5	4	9	1	0	1	0.5556	0.2469
10	4	2	6	0	1	1	0.6667	0.2222
15	3	1	4	1	0	1	0.7500	0.1875
19	2	1	3	0	1	1	0.6667	0.2222
23	1	0	1	1	0	1	1.0000	0.0000
Sum				3			4.1389	1.1288

(The variance at $t = 23$ is 0 because only one subject is at risk; $n_i - 1 = 0$ in the denominator, but $n_{1,i} n_{2,i} = 0$ in the numerator dominates and the contribution is 0.)

The log-rank statistic is

$$Z^2 = \frac{(\sum_i d_{1,i} - \sum_i E(d_{1,i}))^2}{\sum_i \text{Var}(d_{1,i})} = \frac{(3 - 4.1389)^2}{1.1288} = \frac{1.2972}{1.1288} \approx 1.149, \quad Z^2 \sim \chi_1^2.$$

The p-value is $P(\chi_1^2 > 1.149) \approx 0.284$.

Problem 2

Consider the `kidtran` dataset in the `KMsurv` package, where the death times of kidney transplant patients are reported. The patients can be classified by race (1 = white, 2 = black) and gender (1 = male, 2 = female) into one of four groups. Compare the survival curves for different races in each sex group.

Per the question, the analysis is run *separately within each sex* — a two-sample race comparison among males, and another among females. For each sex we plot Kaplan-Meier curves for the two race groups and run a log-rank test.

```
data(kidtran)
str(kidtran)

## 'data.frame':   863 obs. of  6 variables:
## $ obs      : int  1 2 3 4 5 6 7 8 9 10 ...
## $ time     : int  1 5 7 9 13 13 17 20 26 26 ...
## $ delta    : int  0 0 1 0 0 0 1 0 1 1 ...
## $ gender   : int  1 1 1 1 1 1 1 1 1 1 ...
## $ race     : int  1 1 1 1 1 1 1 1 1 1 ...
## $ age      : int  46 51 55 57 45 43 47 65 55 44 ...

male  <- subset(kidtran, gender == 1)
female <- subset(kidtran, gender == 2)
```

Males: KM curves and log-rank by race

```
fit_m <- survfit(Surv(time, delta) ~ race, data = male)
plot(fit_m, mark.time = TRUE, lty = 1, col = c("black", "red"),
     xlab = "Time (days)", ylab = "Survival probability",
     main = "Males: KM by race")
legend("bottomleft", c("Race 1 (white)", "Race 2 (black)"),
     lty = 1, col = c("black", "red"), bty = "n")
```

```
survdif(Surv(time, delta) ~ race, data = male)
```

```
## Call:
## survdiff(formula = Surv(time, delta) ~ race, data = male)
##
##           N Observed Expected (O-E)^2/E (O-E)^2/V
## race=1 432         73      71.9    0.0168    0.097
## race=2  92         14      15.1    0.0801    0.097
##
## Chisq= 0.1  on 1 degrees of freedom, p= 0.8
```

Hypotheses: $H_0: S_{\text{white}}(t) = S_{\text{black}}(t)$ for all t within males vs H_1 : not equal for some t .

Test statistic $\chi^2 = 0.10$ on $df = 1$, $p \approx 0.76$. **Do not reject H_0 .** Among males, there is no evidence of a race difference in survival; the two KM curves track each other closely.

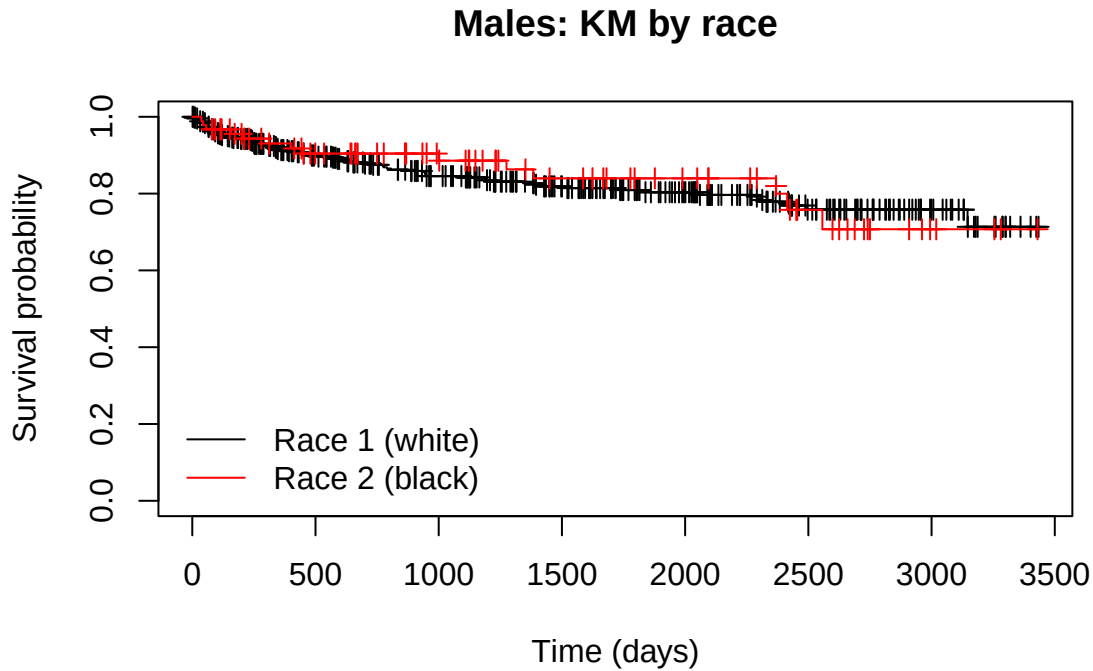


Figure 1: Kaplan-Meier survival by race, males only. Tick marks denote right-censored observations.

Females: KM curves and log-rank by race

```
fit_f <- survfit(Surv(time, delta) ~ race, data = female)
plot(fit_f, mark.time = TRUE, lty = 1, col = c("black", "red"),
     xlab = "Time (days)", ylab = "Survival probability",
     main = "Females: KM by race")
legend("bottomleft", c("Race 1 (white)", "Race 2 (black)"),
     lty = 1, col = c("black", "red"), bty = "n")
```

```
survdif(Surv(time, delta) ~ race, data = female)
```

```
## Call:
## survdiff(formula = Surv(time, delta) ~ race, data = female)
##
##           N Observed Expected (O-E)^2/E (O-E)^2/V
## race=1 280      39    44.79    0.748    4.85
## race=2  59      14     8.21    4.076    4.85
##
## Chisq= 4.8  on 1 degrees of freedom, p= 0.03
```

Hypotheses: $H_0 : S_{\text{white}}(t) = S_{\text{black}}(t)$ for all t within females vs H_1 : not equal for some t .

Test statistic $\chi^2 = 4.8$ on $df = 1$, $p \approx 0.028$. **Reject** H_0 at the 5% level. Among females, survival differs significantly by race: black female recipients had more observed deaths than expected under H_0 (14 vs 8.21), so they have *worse* survival than white female recipients, consistent with the KM plot in which the race-2 curve drops below the race-1 curve.

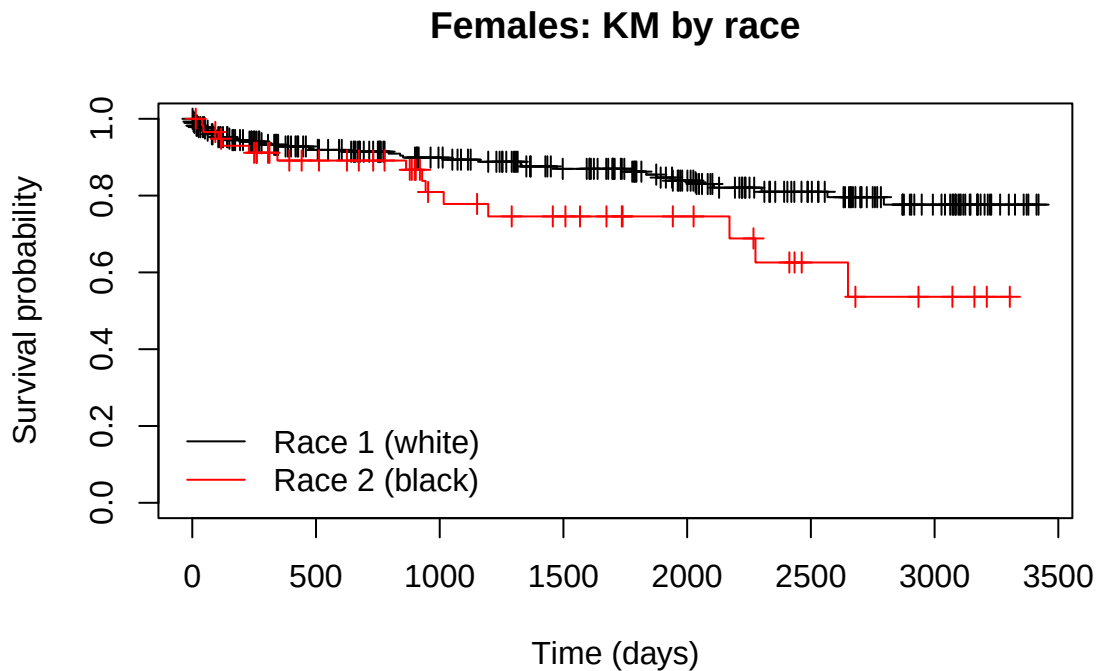


Figure 2: Kaplan-Meier survival by race, females only. Tick marks denote right-censored observations.

Problem 3

Consider the `larynx` data in the `KMsurv` package. Let

- $Z_1 = 1$ if the patient is in stage II, 0 otherwise;
- $Z_2 = 1$ if the patient is in stage III, 0 otherwise;
- $Z_3 = 1$ if the patient is in stage IV, 0 otherwise;
- $Z_4 = \text{age (in years)}$.

Fit a proportional-hazards model with Z_1, Z_2, Z_3, Z_4 and the interaction $Z_1 \cdot Z_4$. Explain the results. What is the relative risk of dying for a stage-II patient of age 50 compared to a stage-I patient of the same age?

Stage I is the reference category.

Cox PH fit

```
data(larynx)
larynx$Z1 <- as.numeric(larynx$stage == 2) # stage II indicator
larynx$Z2 <- as.numeric(larynx$stage == 3) # stage III indicator
larynx$Z3 <- as.numeric(larynx$stage == 4) # stage IV indicator
larynx$Z4 <- larynx$age

fit <- coxph(Surv(time, delta) ~ Z1 + Z2 + Z3 + Z4 + Z1:Z4,
```

```

data = larynx, ties = "breslow")
summary(fit)

## Call:
## coxph(formula = Surv(time, delta) ~ Z1 + Z2 + Z3 + Z4 + Z1:Z4,
##       data = larynx, ties = "breslow")
##
##      n= 90, number of events= 50
##
##              coef exp(coef)    se(coef)      z Pr(>|z|)
## Z1      -7.3820143  0.0006223   3.4027542 -2.169   0.0301 *
## Z2       0.6218044  1.8622853   0.3558078  1.748   0.0805 .
## Z3       1.7534270  5.7743576   0.4239595  4.136 3.54e-05 ***
## Z4       0.0059729  1.0059908   0.0148792  0.401   0.6881
## Z1:Z4    0.1116674  1.1181409   0.0476728  2.342   0.0192 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## Z1      0.0006223  1606.8231 7.900e-07   0.4903
## Z2      1.8622853    0.5370 9.272e-01   3.7403
## Z3      5.7743576    0.1732 2.516e+00  13.2550
## Z4      1.0059908    0.9940 9.771e-01   1.0358
## Z1:Z4    1.1181409    0.8943 1.018e+00   1.2277
##
## Concordance= 0.682  (se = 0.04 )
## Likelihood ratio test= 24.11  on 5 df,   p=2e-04
## Wald test               = 23.77  on 5 df,   p=2e-04
## Score (logrank) test = 27.98  on 5 df,   p=4e-05

```

The fitted model is

$$h(t | Z) = h_0(t) \exp\{\beta_1 Z_1 + \beta_2 Z_2 + \beta_3 Z_3 + \beta_4 Z_4 + \beta_{14} Z_1 Z_4\}.$$

Interpretation

Recall that hazard ratios are constant in time under the PH assumption (Lec 23, slide 6) but here vary across covariate combinations because of the interaction. From the coefficient table:

- $\hat{\beta}_2 = 0.622$, $\exp(\hat{\beta}_2) = 1.86$ ($p = 0.08$). Stage III versus stage I patients have 1.86 times the hazard of death at *any* age (no interaction with age was fit for stage III).
- $\hat{\beta}_3 = 1.753$, $\exp(\hat{\beta}_3) = 5.77$ ($p < 0.001$). Stage IV versus stage I patients have 5.77 times the hazard at any age — stage IV is highly significantly worse than stage I.
- $\hat{\beta}_4 = 0.0060$, $\exp(\hat{\beta}_4) = 1.006$ ($p = 0.69$). Because $Z_1 \cdot Z_4$ is in the model, β_4 is the per-year log hazard ratio for *non-stage-II* patients (stages I, III, IV); a one-year increase in age multiplies their hazard by 1.006, not significantly different from 1.
- $\hat{\beta}_1 = -7.382$, $\exp(\hat{\beta}_1) = 6.2 \times 10^{-4}$ ($p = 0.03$). This is *not* the stage II vs stage I hazard ratio in general; it is the HR at age 0, which is far outside the range of the data. Because of the $Z_1 \cdot Z_4$ interaction, the stage II vs stage I log hazard ratio at age a equals $\beta_1 + a \beta_{14}$.
- $\hat{\beta}_{14} = 0.1117$, $\exp(\hat{\beta}_{14}) = 1.118$ ($p = 0.02$). Each additional year of age multiplies the stage-II-vs-stage-I hazard ratio by 1.12. Equivalently, for stage II patients, the per-year log hazard ratio is $\beta_4 + \beta_{14} = 0.118$, so a one-year increase in age multiplies a stage-II patient's hazard by $\exp(0.118) \approx 1.12$. The age effect

is therefore much stronger in stage II than in the other stages, and the stage-II-vs-stage-I gap *widens* with increasing age.

The omnibus likelihood-ratio test (24.1 on 5 df, $p = 2 \times 10^{-4}$) indicates that the covariate set is jointly highly significant.

Relative risk: stage II at age 50 vs stage I at age 50

Both patients have $Z_2 = Z_3 = 0$, $Z_4 = 50$. Their covariate vectors are

- Stage I: $(Z_1, Z_2, Z_3, Z_4, Z_1 Z_4) = (0, 0, 0, 50, 0)$;
- Stage II: $(Z_1, Z_2, Z_3, Z_4, Z_1 Z_4) = (1, 0, 0, 50, 50)$.

The log hazard ratio is therefore

$$\log \text{RR} = \beta_1 (1 - 0) + \beta_{14} (50 - 0) = \beta_1 + 50 \beta_{14}.$$

Plugging in the estimates,

$$\widehat{\log \text{RR}} = -7.3820 + 50 (0.1117) = -1.7986,$$

$$\widehat{\text{RR}} = \exp(-1.7986) \approx 0.166.$$

```
b1 <- coef(fit)["Z1"]
b14 <- coef(fit)["Z1:Z4"]
logRR <- b1 + 50 * b14
RR <- exp(logRR)
c(logRR = unname(logRR), RR = unname(RR))
```

```
##      logRR      RR
## -1.7986446  0.1655231
```

Answer. A stage II patient aged 50 has an estimated hazard about $\widehat{\text{RR}} \approx 0.17$ times that of a stage I patient aged 50 — i.e., at age 50 a stage II patient has *lower* mortality hazard than a stage I patient. (Because $\hat{\beta}_{14} > 0$, this gap shrinks at older ages and reverses around age $\beta_1/(-\beta_{14}) \approx 66$ years, beyond which stage II patients have higher hazard than stage I patients.)